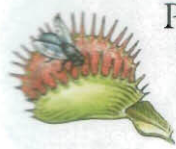


CARNIVOROUS PLANTS

PLANTS THAT catch and “eat” insects are called carnivorous plants. These plants fall into two groups.

Some species, such as the Venus flytrap, have active traps with moving parts. Other species have passive traps, catching their victims on a sticky surface or drowning them in a pool of fluid. Carnivorous plants live in areas where the soil is poor in nitrates and other nutrients, such as bogs, peatlands, and swamps. They obtain extra nutrients by catching insects, which are digested by special juices.



Monkey-cup pitcher plant



Tendrils

Pitcher plants from Southeast Asia form traps that hang from their leaves.

Hanging pitcher

Passive traps

Most carnivorous plants have passive traps. Usually the leaves of these plants have evolved to catch insects in a variety of ways. Some are sticky, others form pit-fall traps with fluid at the bottom and are called pitcher plants.

The lid and the smooth rim are often brightly coloured to attract insects.

Rim of the pitcher contains nectar.

Mouth of pitcher

The lid stays closed while the pitcher develops.

Development of a pitcher plant

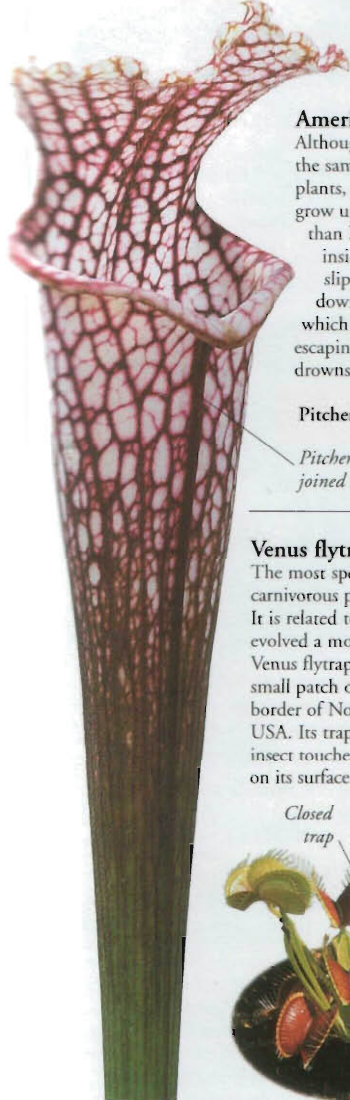
1 A young leaf tip extends into a tendril.

2 An upturned swelling appears at the end.

3 The swelling develops into a pitcher.

4 The lid opens when the pitcher is mature.

Insects fall into the liquid and are digested.



American pitcher plants

Although they catch their prey in the same way as other pitcher plants, American pitcher plants grow up from the ground rather than hanging from leaves. The inside of the pitcher is slippery and lined with downward pointing hairs which prevent the insects from escaping. The liquid below drowns and slowly digests them.

Pitcher plant

Pitcher is made of leaves joined at the edges.

Venus flytrap

The most spectacular of all the carnivorous plants is the Venus flytrap. It is related to the sundews but has evolved a more elaborate trap. The Venus flytrap grows wild only in one small patch of marshy ground on the border of North and South Carolina, USA. Its trap springs closed when an insect touches the hairs on its surface.



Closed trap

Stimulation of at least three trigger hairs sets off the mechanism that closes the trap.

Surface of the trap

Venus flytrap

Magnified view of a trigger hair

Butterworts

These small plants have sticky leaves. Small flies are attracted to their smell and get stuck. The leaves slowly roll up, and the insects are digested by juices that ooze out of the leaf.



Common butterwort

Leaf

Sundew flowers develop at the end of a long stalk.



A fly stuck to the hairs on a sundew leaf

Cape sundew

Active traps

Any trap with moving parts is called an active trap. These include plants such as sundews and butterworts, and the Venus flytrap.

Sundews

The upper surface of a sundew leaf is covered with red hairs that secrete drops of clear, sticky liquid. Insects get stuck, then the edges of the leaf slowly roll inward enclosing the insect, and the plant secretes juices that digest it.



Sticky leaf

Bladderworts

These are rootless water plants. Their leaves and stems bear tiny bladders with a lid covered in sensitive hairs. If a creature brushes the hairs, the lid of the bladder flips open. Water rushes in, carrying the victim with it.



Greater bladderwort

How a Venus flytrap works

1 An insect lands on a leaf, touching the sensitive trigger hairs.

2 The leaf closes, and the spines interlock, trapping the insect.

3 The trap is fully closed in 30 minutes, and digestion begins.



Trigger hair

Trap is fringed with long spines.



Insect is trapped in one-fifth of a second.



FIND OUT MORE

ASIAN WILDLIFE

FLOWERS

INSECTS

NORTH AMERICAN WILDLIFE

PLANTS

PLANTS, ANATOMY

PLANTS, DEFENCE

PLANTS, REPRODUCTION

SOUTH AMERICAN WILDLIFE